

When Is the Mean Nearly Normal?

AP Statistics · Unit 2 · Topic 2.12 · B: 2026-11-19 · E: 2026-11-20 · TEACHER KEY

CED TETHER

- **Skill 3.C** — Describe probability distributions.
- **Enduring Understanding UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.H** — Estimate sampling distributions using simulation. [Skill 3.C]
- **EK UNC-3.H.1** — A sampling distribution of a statistic is the distribution of values for the statistic for all possible samples of a given size from a given population.
- **EK UNC-3.H.2** — The central limit theorem (CLT) states that when the sample size is sufficiently large, a sampling distribution of the mean of a random variable will be approximately normally distributed.
- **EK UNC-3.H.3** — The central limit theorem requires that the sample values are independent of each other and that n is sufficiently large.
- **EK UNC-3.H.4** — A randomization distribution is a collection of statistics generated by simulation assuming known values for the parameters.
- **EK UNC-3.H.5** — The sampling distribution of a statistic can be simulated by generating repeated random samples from a population.

Essential question: How do population shape and sample size determine whether a normal model is trustworthy for sample means?

DOK-3 skill: 4.B · **FRQ pattern:** clt-when-does-normality-apply

DOK rationale: Part (c) weighs population shape, independence, and sample size to judge two normality claims and distinguish a model for sample means from a model for individuals.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Board slide up at the bell. Ask for a sample-size choice before displaying a rule of thumb.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board slide at minute 33. Circulate and require students to name which distribution becomes approximately normal.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose n equals 4 or n equals 40 and cite population shape or sample size.
- Use the printed formulas to find the sampling mean and standard deviation for each sample size.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Does averaging four values erase a long right tail automatically?
- Which distribution does the CLT describe: commuters or sample means?
- What happens to $15/\sqrt{n}$ when n grows from 4 to 40?

- Which probability calculation becomes questionable if the n equals 4 shape is not normal?

Adult role

Solo teacher. Circulate 5 pairs. Require shape, sample size, and independence in every normality defense; redirect claims that the population itself changes shape.

[WARN] LOOK FOR

- Invoking the CLT for n equals 4 without considering the strong population skew.
- Saying forty individual commute times have a normal distribution.
- Using 15 minutes, rather than $15/\sqrt{n}$, for the spread of sample means.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Rejects automatic normality at $n=4$ using population right skew.
- **judgment-2** (required) — Supports $n=40$ as more defensible using independence and the CLT, without claiming a universal guarantee.
- **judgment-3** (required) — Distinguishes sample means from individual times and explains why a small-sample normal tail can be unreliable.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

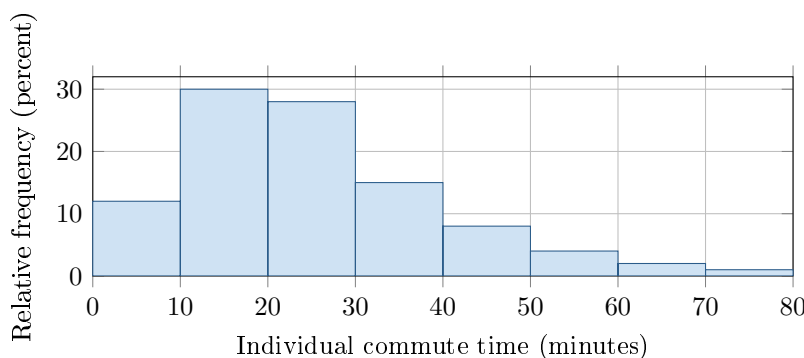
- Treating the central limit theorem as applying at every sample size
- Saying the individual commute-time population becomes normal when n increases
- Dividing the mean, rather than the standard deviation, by $\sqrt{n}$
- Using the number of repeated samples instead of the observations per sample as n

[STAR] TODAY'S PROBLEM — annotated

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: “Any average is approximately normal, even for $n = 4$.”

Sol: “With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible.”



FIRST TAKE (5 min)

Which would you expect to smooth out the long commute times more: averaging 4 people or 40? Give one reason.

Key: Not graded. Expect ‘averages are normal’ as an initial rule; the revision should add population shape and sufficiently large n .

Rules callout “CLT: SHAPE, SIZE, AND THE RIGHT DISTRIBUTION (keep on the page)” is printed on the student sheet.

DOK 1 (a) Identify the population shape and state the population mean and standard deviation with units.

Key: The population distribution of individual commute times is strongly right-skewed. Its mean is $\mu = 25$ minutes and its standard deviation is $\sigma = 15$ minutes.

DOK 2 (b) For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.

Key: For either sample size, $\mu_{\bar{x}} = \mu = 25$ minutes. For $n = 4$, $\sigma_{\bar{x}} = 15/\sqrt{4} = 7.5$ minutes. For $n = 40$, $\sigma_{\bar{x}} = 15/\sqrt{40} \approx 2.3717$ minutes, or about 2.37 minutes.

DOK 3 (c) In 3–4 sentences, decide whose claim is better supported. Use the population shape, independence, and the two sample sizes. Explain which distribution the CLT describes and why using a normal curve for averages of 4 could be unreliable.

Key: Sol’s claim is better supported: a long right tail can remain when only four commute times are averaged. Independent samples of 40 make an approximately normal distribution of sample means more defensible by the CLT. The CLT describes sample means, not individual commute times. Using a normal curve for averages of 4 could give unreliable tail probabilities.

EXIT

One thing the rules box changed about my first take: _____